

Topic 7

Transformations and vectors

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

Learning focus

Explain every method before practising. Use the original practice questions to check calculation, reasoning and communication.

Topic map

7.1	Translations and reflections
7.2	Rotations and enlargements
7.3	Combined transformations
7.4	Vectors and magnitude
7.5	Vector geometry and position

Core foundations and Extended applications are integrated throughout. Study each example, then attempt the practice paper without the notes.

1. Moving a shape

A translation moves each point by the same column vector, such as (3,-2): right 3 and down 2. A reflection uses a mirror line and reverses orientation. A rotation requires its centre, angle and direction. Give a transformation in full to distinguish motions with similar images.

Worked example

Translate A(1,4) by (3,-2): $A' = (1+3, 4-2) = (4,2)$.

Exam check: A rotation answer needs the centre as well as its direction and angle.

2. Enlargement and scale

An enlargement with scale factor k sends a point X from centre O to X' with $OX' = k OX$ as directed. Negative scale factors place images across the centre. Lengths change by $|k|$ and areas by k^2 . Use lines through corresponding vertices to locate the centre.

Worked example

Centre origin, $k = -2$: (3,1) maps to (-6,-2). An original area of 5 cm^2 becomes 20 cm^2 .

Exam check: A negative scale factor is not a reflection in an arbitrary axis.

3. Adding and scaling vectors

Vectors carry magnitude and direction. Add vectors head-to-tail; in component form add matching coordinates. $A - B$ means $A + (-B)$. Multiply a vector by a scalar to change its length and possibly reverse its direction. Magnitude of (a,b) is $\sqrt{a^2+b^2}$.

Worked example

$(2,-3) + (-5,4) = (-3,1)$; magnitude of (3,4) is 5.

Exam check: Keep x and y components in their own columns.

4. Position vectors and geometry

A position vector runs from the origin to a point. If $OA = a$ and $OB = b$, then $AB = b - a$. The midpoint M of AB has position vector $(a+b)/2$. For collinearity, show vectors are scalar multiples; for parallel lines, show their direction vectors are multiples.

Worked example

If $a = (1,2)$ and $b = (7,8)$, then $AB = (6,6)$, and $OM = (4,5)$.

Exam check: AB and BA have opposite signs; subtract endpoint minus starting point.

5. Combining transformations

Order matters: translate then reflect can yield a different answer from reflect then translate. Apply each operation to the original vertices one at a time, then label the intermediate image. A coordinate grid and matching arrows make the sequence easier to check.

Worked example

Start with $(2,1)$; reflect in y -axis to $(-2,1)$, then translate by $(3,-2)$ to $(1,-1)$.

Exam check: Do not combine two transformations before checking their order.

Before the practice paper

Method and accuracy

Write the formula or first operation, show the intermediate step, give units when needed and check by substitution, estimation or a sketch. On non-calculator work, keep exact forms when requested.

Practice paper

Topic 7 · Transformations and vectors

Instructions

Original Cambridge-style practice. Answer all questions. Section A is multiple choice; Section B requires working and final answers. Try Section A without a calculator; use a calculator in Section B only where needed.

Candidate name:	
Class / date:	

Total: 30 marks

Section A · Multiple choice

Choose one answer for each question.

1. Translate (2,3) by (4,-1). [1]

- A (6,2)
- B (-2,4)
- C (8,3)
- D (2,-1)

2. Reflect (3,-2) in the x-axis. [1]

- A (-3,-2)
- B (-3,2)
- C (3,2)
- D (2,-3)

3. Rotate (2,1) 180° about origin. [1]

- A (1,2)
- B (-2,-1)
- C (2,-1)
- D (-1,2)

4. Enlarge (1,-3) by scale factor 2 about origin. [1]

- A (2,-6)
- B (-2,6)
- C (1,-6)
- D (3,-1)

5. Find $(2,5) + (-3,4)$. [1]

- A (5,9)
- B (-1,9)
- C (-1,1)
- D (1,20)

6. Magnitude of (6,8)? [1]

- A 7
- B 10
- C 14
- D 100

7. If $OA = a$ and $OB = b$, AB equals [1]

- A $a+b$
- B $a-b$
- C $b-a$
- D $-a-b$

8. For length scale factor -3, area factor is [1]

- A -9
- B -3
- C 3
- D 9

Section B · Structured questions

9. Triangle $A(1,1)$, $B(3,1)$, $C(1,4)$ is translated by $(4,-2)$.

(a) Give the coordinates of A' . [1]

(b) Give the coordinates of B' and C' . [2]

(c) Describe what happens to its area. [2]

[Total: 5]

10. Point $P(2,-1)$ is enlarged about the origin with scale factor -3 .

(a) Find the image P' . [2]

(b) Find OP and OP' . [2]

(c) State the factor by which area changes. [1]

[Total: 5]

11. Position vectors $OA = (2,3)$ and $OB = (8,11)$.

(a) Find AB . [2]

(b) Find the midpoint M of AB . [2]

(c) Find the magnitude of AB . [2]

[Total: 6]

12. Start with point $X(4,2)$. Reflect it in the y -axis, then translate by $(3,-5)$.

(a) Find the reflected point $X^\blacksquare$. [1]

(b) Find the final point $X^\blacksquare$. [2]

(c) Apply these two transformations in reverse order and find the final point.

[3]

[Total: 6]

Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Award method marks when the mathematical setup is correct even if an arithmetic slip follows.
Check each part independently.

Section A · Answers and reasons

1. A — (6,2). Add the translation vector component by component.
2. C — (3,2). Reflection in the x-axis changes the sign of y alone.
3. B — (-2,-1). A half-turn about the origin negates both coordinates.
4. A — (2,-6). Multiply both coordinates by 2 about the origin.
5. B — (-1,9). Add x-components and y-components separately.
6. B — 10. Magnitude is $\sqrt{6^2+8^2} = 10$.
7. C — b-a. Subtract the starting position from the endpoint: b - a.
8. D — 9. Area scales by the square of -3, which is 9.

Section B · Worked solutions

9. a) $(1+4, 1-2) = (5, -1)$ [1]. b) $B' = (7, -1)$, $C' = (5, 2)$ [2]. c) Translation preserves all lengths and area, so the area is unchanged [2].

10. a) $-3(2, -1) = (-6, 3)$ [2]. b) $OP = \sqrt{5}$ and $OP' = \sqrt{45} = 3\sqrt{5}$ [2]. c) $(-3)^2 = 9$ [1].

11. a) $b-a = (8-2, 11-3) = (6, 8)$ [2]. b) $((2+8)/2, (3+11)/2) = (5, 7)$ [2]. c) $\sqrt{6^2+8^2} = 10$ [2].

12. a) $(-4, 2)$ [1]. b) $(-4+3, 2-5) = (-1, -3)$ [2]. c) Translate first: $(7, -3)$; reflect: $(-7, -3)$ [3].

Review your errors

Mark the first wrong step: setup, algebra, arithmetic, units, graph or reasoning. Redo that part without seeing the solution.

Syllabus reference: Cambridge IGCSE Mathematics 0580, 2025-2027,
www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf. Paper style informed by the June 2026 papers and examiner report provided by the user. All practice questions are original.